

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

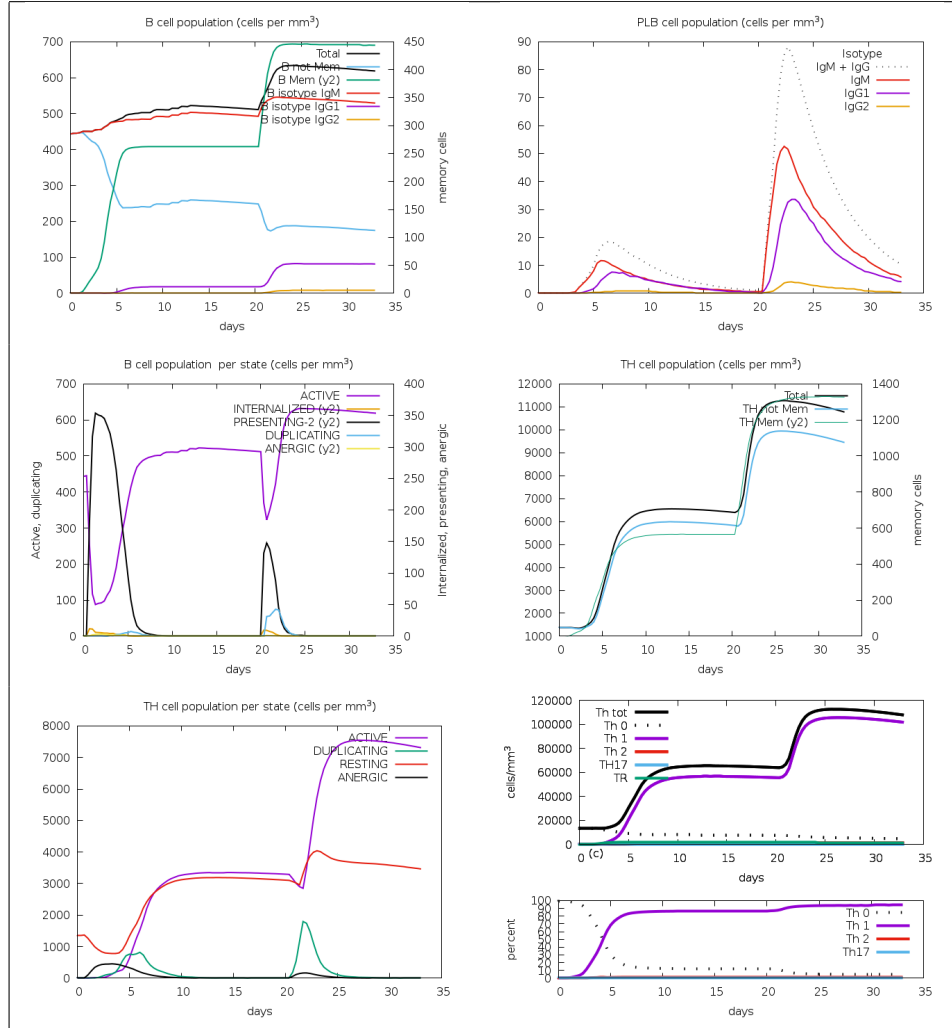


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

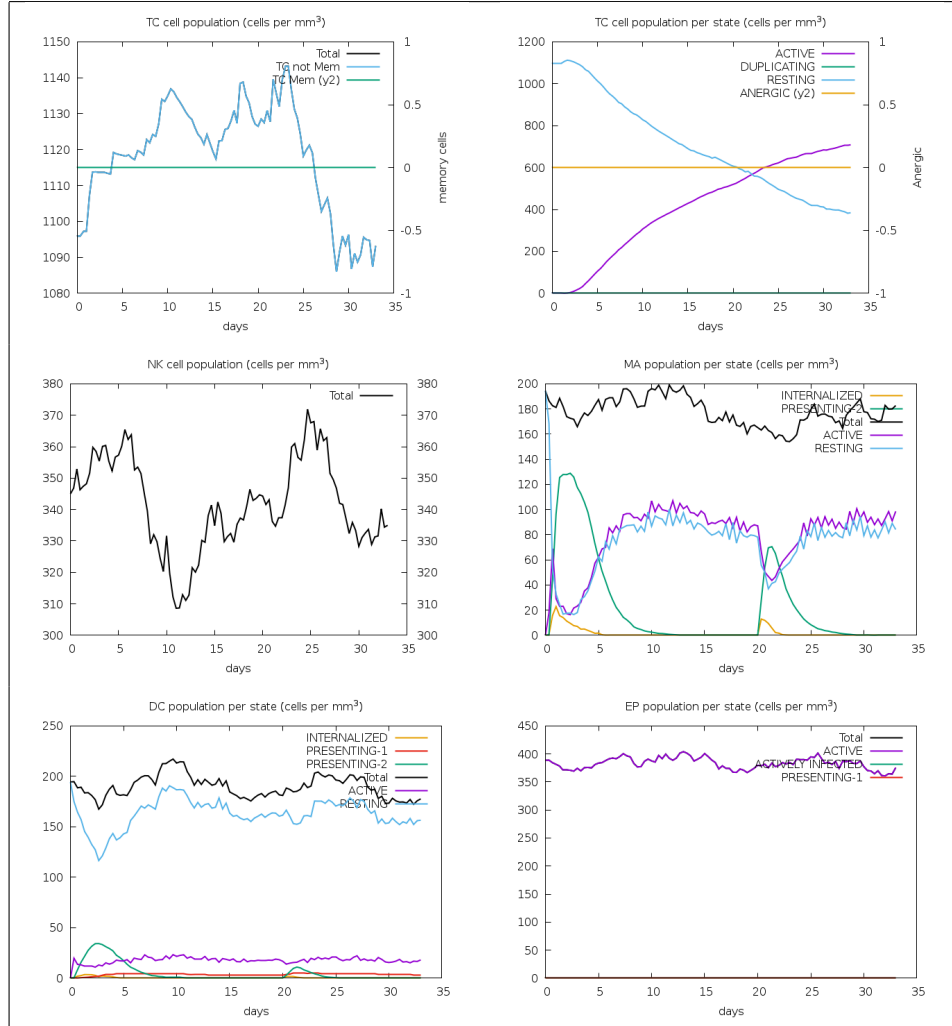


Figure 2: Legend: symbols as figure above.

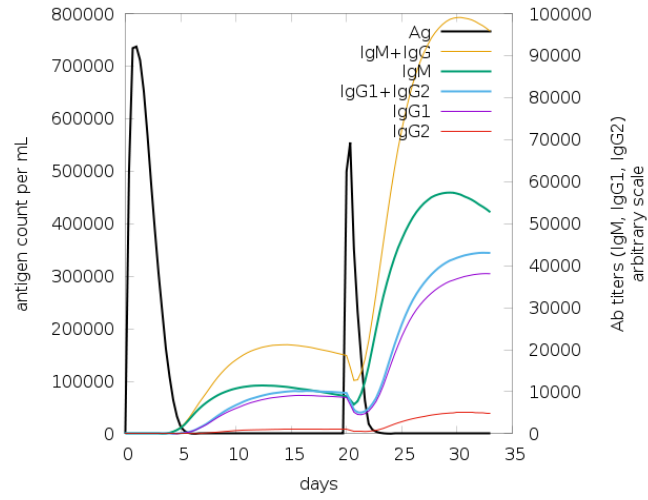


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

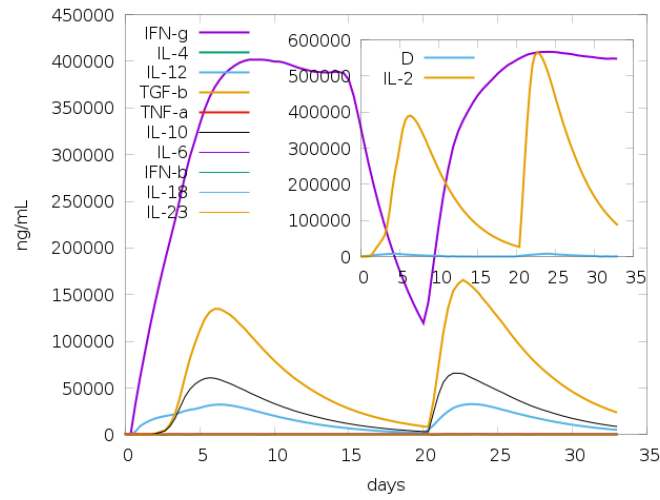


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Parker_B_scale_epitope_find
Use Parker's propensity scale, takes an antigen block as input,
and creates a list of residues that are possible epitopes.

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013846_5_cG22o6ChAcL0.FSA_1_001

AKFVAAWTLKAAAGGSPGIDELAQQHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNLSILFALLAAAAAANFKSLSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQQNWAAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPLSAAAAA
ASCFTASVSTVVSAT

0000000000001010110000000111111100000000000011000000001101000000000000000000
000000001000
00
0000000000000000

Epitopes of protein 0 -----
1] pos=25 len=7 HASGEGP

DoPeptideList_I:
Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

=====

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013846_5_cG22o6ChAcL0.FSA_1_001

AKFVAAWTLKAAAGGSPGIDELAQQHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNLSILFALLAAAAAANFKSLSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQQNWAAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 0 -----
0] pos= 43 score=0.030800 unnormalised=3.3916000000 AADQATRVY
1] pos= 171 score=0.002058 unnormalised=0.2266000000 AAAAAIALY
2] pos= 198 score=0.015053 unnormalised=1.6576000000 YAAAAADPY
3] pos= -1 score=0.952089 unnormalised=104.8410000000 non-binding event

=====

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013846_5_cG22o6ChAcL0.FSA_1_001

AKFVAAWTLKAAAGGSPGIDELAQQHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQQNWAAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 0 -----

0]	pos=	43	score=0.030800	unnormalised=3.3916000000	AADQATRVY
1]	pos=	171	score=0.002058	unnormalised=0.2266000000	AAAAAIALY
2]	pos=	198	score=0.015053	unnormalised=1.6576000000	YAAAAADPY
3]	pos=	-1	score=0.952089	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702
Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013846_5_cG22o6ChAcL0.FSA_1_001

AKFVAAWTLKAAAGGSPGIDELAQQHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQQNWAAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 0 -----

0]	pos=	34	score=0.035294	unnormalised=4.5212000000	NPISKTA AA
1]	pos=	56	score=0.038627	unnormalised=4.9482000000	GPSQAPAAA
2]	pos=	72	score=0.031742	unnormalised=4.0662000000	APQLFAAAA
3]	pos=	126	score=0.019517	unnormalised=2.5002000000	IPNLFAAAA
4]	pos=	170	score=0.002421	unnormalised=0.3102000000	IAAAAAIAL
5]	pos=	208	score=0.025395	unnormalised=3.2532000000	VPFVAAAAA
6]	pos=	227	score=0.019439	unnormalised=2.4902000000	PPRGPLSA
7]	pos=	230	score=0.009150	unnormalised=1.1722000000	GPPLSAAAA
8]	pos=	-1	score=0.818414	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702
Pseudo sequence: KAAREEQQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013846_5_cG22o6ChAcL0.FSA_1_001

AKFVAAWTLKAAAGGSPGIDELAQQHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQQNWAAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 0 -----

0]	pos=	34	score=0.035294	unnormalised=4.5212000000	NPISKTA AA
1]	pos=	56	score=0.038627	unnormalised=4.9482000000	GPSQAPAAA
2]	pos=	72	score=0.031742	unnormalised=4.0662000000	APQLFAAAA

3]	pos= 126	score=0.019517	unnormalised=2.5002000000	IPNLFAAAA
4]	pos= 170	score=0.002421	unnormalised=0.3102000000	IAAAAAIAL
5]	pos= 208	score=0.025395	unnormalised=3.2532000000	VPFVAAAAA
6]	pos= 227	score=0.019439	unnormalised=2.4902000000	PPRGPPLSA
7]	pos= 230	score=0.009150	unnormalised=1.1722000000	GPPLSAAAA
8]	pos= -1	score=0.818414	unnormalised=104.8410000000	non-binding event

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

=====

Allele: DRB1_0101
Pseudo sequence: KAFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013846_5_cG22o6ChAcL0.FSA_1_001

AKFVAAWTLKAAAGGSPGIDELAQHASGEGPGINPISKTA AAAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNSILFALLAAAAAANFKSLSTEFIPNLFAAAAAANSTNITAVVRAQGGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQKNWAAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 0 -----

0]	pos= 2	score=0.013245	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001806	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 36	score=0.021107	unnormalised=5.3935600000	ISKTA AAAA
3]	pos= 51	score=0.012658	unnormalised=3.2345600000	YQDLGGPSQ
4]	pos= 67	score=0.033159	unnormalised=8.4735600000	YQAYAAPQL
5]	pos= 75	score=0.023728	unnormalised=6.0635600000	LFAAAAAGT
6]	pos= 76	score=0.008823	unnormalised=2.2545600000	FAAAAAAGTY
7]	pos= 84	score=0.008193	unnormalised=2.0935600000	YKRVTEKAA
8]	pos= 99	score=0.016798	unnormalised=4.2925600000	FNSILFALL
9]	pos= 103	score=0.039836	unnormalised=10.1795600000	LFALLAAAA
10]	pos= 104	score=0.058823	unnormalised=15.0315600000	FALLAAAAA
11]	pos= 106	score=0.037437	unnormalised=9.5665600000	LLAAAAA AA
12]	pos= 107	score=0.035241	unnormalised=9.0055600000	LAAAAA AA
13]	pos= 117	score=0.030651	unnormalised=7.8325600000	FKSLSTEF
14]	pos= 126	score=0.000840	unnormalised=0.2145600000	IPNLFAAAA
15]	pos= 129	score=0.021917	unnormalised=5.6005600000	LFAAAAANS
16]	pos= 130	score=0.052233	unnormalised=13.3475600000	FAAAAAANST
17]	pos= 143	score=0.002201	unnormalised=0.5625600000	VVRAQGLDA
18]	pos= 163	score=0.007860	unnormalised=2.0085600000	VTRNLNAIA
19]	pos= 167	score=0.044727	unnormalised=11.4295600000	LNAIAAAAA
20]	pos= 170	score=0.030585	unnormalised=7.8155600000	IAAAAAIAL
21]	pos= 178	score=0.017041	unnormalised=4.3545600000	LYLQQNWAA
22]	pos= 180	score=0.023764	unnormalised=6.0725600000	LQQNWAAAA
23]	pos= 184	score=0.042133	unnormalised=10.7665600000	WAAAAAATV
24]	pos= 192	score=0.039616	unnormalised=10.1235600000	VKGMQSYAA
25]	pos= 195	score=0.018575	unnormalised=4.7465600000	MQSYAAAAA
26]	pos= 198	score=0.007269	unnormalised=1.8575600000	YAAAAADPY
27]	pos= 210	score=0.024394	unnormalised=6.2335600000	FVAAAAALE
28]	pos= 233	score=0.036509	unnormalised=9.3295600000	LSAAAAAAS

29]	pos= 243	score=0.029900	unnormalised=7.6405600000	FTASVSTVV
30]	pos= -1	score=0.258934	unnormalised=66.1680000000	non-binding event

=====

Allele: DRB1_0101
Pseudo sequence: KFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-013846_5_cG22o6ChAcL0.FSA_1_001

AKFVAAWTLKAAAGGSPGIDELAQHASGEGPGINPISKTAADQATRVYQDLGGPSQAPAAAAAYQAYAAPQLFAAA
AAGTYKRVTEKAAAAADPPFNLSILFALLAAAAAANFKSLSSSTEFIPNLFAAAAANSTNITAVVRAQGLDAAAAATPDI
LPVVTRNLNAIAAAAAIALYLQQNWAAAAATVKGMQSYAAAAADPYQVPFVAAAAALERGPPGPRRPPRGPPLSAAAAA
ASCFTASVSTVVSAT

Epitopes of protein 0 -----

0]	pos= 2	score=0.013245	unnormalised=3.3845600000	FVAAWTLKA
1]	pos= 6	score=0.001806	unnormalised=0.4615600000	WTLKAAAGG
2]	pos= 36	score=0.021107	unnormalised=5.3935600000	ISKTA AAAA
3]	pos= 51	score=0.012658	unnormalised=3.2345600000	YQDLGGPSQ
4]	pos= 67	score=0.033159	unnormalised=8.4735600000	YQAYAAPQL
5]	pos= 75	score=0.023728	unnormalised=6.0635600000	LFAAAAAGT
6]	pos= 76	score=0.008823	unnormalised=2.2545600000	FAAAAAAGTY
7]	pos= 84	score=0.008193	unnormalised=2.0935600000	YKRVTEKAA
8]	pos= 99	score=0.016798	unnormalised=4.2925600000	FNSILFALL
9]	pos= 103	score=0.039836	unnormalised=10.1795600000	LFALLAAAA
10]	pos= 104	score=0.058823	unnormalised=15.0315600000	FALLAAAAA
11]	pos= 106	score=0.037437	unnormalised=9.5665600000	LLAAAAAAA
12]	pos= 107	score=0.035241	unnormalised=9.0055600000	LAAAAAAA
13]	pos= 117	score=0.030651	unnormalised=7.8325600000	FKSLSSSTEF
14]	pos= 126	score=0.000840	unnormalised=0.2145600000	IPNLFAAAA
15]	pos= 129	score=0.021917	unnormalised=5.6005600000	LFAAAAANS
16]	pos= 130	score=0.052233	unnormalised=13.3475600000	FAAAAAANST
17]	pos= 143	score=0.002201	unnormalised=0.5625600000	VVRAQGLDA
18]	pos= 163	score=0.007860	unnormalised=2.0085600000	VTRNLNAIA
19]	pos= 167	score=0.044727	unnormalised=11.4295600000	LNAIAAAAA
20]	pos= 170	score=0.030585	unnormalised=7.8155600000	IAAAAAIAL
21]	pos= 178	score=0.017041	unnormalised=4.3545600000	LYLQQNWAA
22]	pos= 180	score=0.023764	unnormalised=6.0725600000	LQQNWAAAA
23]	pos= 184	score=0.042133	unnormalised=10.7665600000	WAAAAAATV
24]	pos= 192	score=0.039616	unnormalised=10.1235600000	VKGMQSYAA
25]	pos= 195	score=0.018575	unnormalised=4.7465600000	MQSYAAAAA
26]	pos= 198	score=0.007269	unnormalised=1.8575600000	YAAAAADPY
27]	pos= 210	score=0.024394	unnormalised=6.2335600000	FVAAAAALE
28]	pos= 233	score=0.036509	unnormalised=9.3295600000	LSAAAAAAS
29]	pos= 243	score=0.029900	unnormalised=7.6405600000	FTASVSTVV
30]	pos= -1	score=0.258934	unnormalised=66.1680000000	non-binding event